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Numerical instability of algebraic rootfinders

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ABSTRACT

We demonstrate that the most popular variants of all common algebraic multidimensional rootfinding algorithms are unstable by analyzing the conditioning of subproblems that are constructed at intermediate steps. In particular, we give multidimensional polynomial systems for which the conditioning of a subproblem can be worse than the conditioning of the original problem by a factor that grows exponentially with the number of variables.

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1. Introduction

For over a decade, the scientific community has been searching for a numerically robust multivariate polynomial rootfinder by recasting the problem as an eigenproblem [1–9]. We are yet to find one. This paper demonstrates that the most popular variants of all common algebraic multidimensional rootfinding algorithms are exponentially unsta-

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ble. This includes hidden-variable resultants (see [2]), Gröbner bases (see section 3), the rational univariate representation (see section 4), multiparameter eigenvalue problems (see section 5), normal form methods (see section 6), and Macaulay resultants (see section 7). We believe that we need a new way to recast multivariate rootfinding problems into eigenproblems that avoids exponential instability.

There is also a large class of global rootfinders, that we do not consider here, based on homotopy continuation [10–13]. Homotopy methods have strong local stability and convergence properties [14,15], but, globally, they can experience arbitrary instability when distinct paths approach each other; this occurs with low probability, and is overcome by randomization and the use of adaptive multiprecision [12,16,17]. A more thorough analysis of the stability of homotopy methods is out of the scope of this paper.

A multivariate rootfinder computes all the solutions of a polynomial system of the form:

$$\begin{pmatrix} p_1(x_1,\ldots,x_d) \\ \vdots \\ p_d(x_1,\ldots,x_d) \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}, \quad (x_1,\ldots,x_d) \in \mathbb{C}^d, \tag{1.1}$$

where $d \geq 2$, and $p_1,\ldots,p_d$ are polynomials in $x_1,\ldots,x_d$ with complex coefficients. We assume that the system in eq. (1.1) has a finite number of simple roots with no roots at infinity. A multivariate rootfinder is a global rootfinder aimed at not missing any solutions to eq. (1.1). Global rootfinders differ from local ones such as Newton–Raphson, as they can find all the equilibrium points of a dynamical system or compute the global maximum of a polynomial.

Once we have a rootfinder that does not miss any solutions, we want the computed solutions to be accurate. Since, at some stage of a rootfinder, there must be a numerical calculation, as proved by the Abel–Ruffini theorem [18,19], the scientific community focuses on well-conditioned problems. For example, consider Wilkinson's polynomial [20] given by $w(x) = (x-1)(x-2)\cdots(x-20)$. If we increase the coefficient of x^{19} by $2^{-23} \approx .0000001$, the root at $x = 20$ moves to $x \approx 20.8$. Computing the roots of $w(x)$ given its coefficients is an ill-conditioned problem, i.e., small perturbations of its coefficients can cause large changes to its roots. We are not interested in computing the roots of $w(x)$ as it is numerically a frivolous task [20]. Instead, in scientific computing, we mainly focus on well-conditioned problems.

1.1. Motivation for eigenvalue-based approaches

Given a well-conditioned rootfinding problem, we would like to derive a stable algorithm to solve it. Roughly speaking, an algorithm is stable if it computes an accurate solution to well-conditioned problems (see section 2.2). The search for a stable algorithm for multivariate polynomial rootfinding is motivated by the existence of stable algorithms for many related problems (see Table 1). All the univariate problems, such as

Table 1
Types of linear equations, rootfinding, and eigenproblems.

Univariate Problems		
	Scalar Problems	Matrix Problems
Linear Problems	Basic Algebra $ax = b$	Generalized Eigenproblem $Ax = \lambda Bx$
Polynomial Problems	Polynomial Rootfinding $p(x) = 0$	Polynomial Eigenproblem $P(x)v = 0$
Multivariate Problems		
	Scalar Problems	Matrix Problems
Linear Problems	Linear System $A\mathbf{x} = \mathbf{b}$	Multivariate Eigenproblem $W_i(\mathbf{x})\mathbf{v}_i = 0, 1 \le i \le d$
Polynomial Problems	Multivariate Polynomial Rootfinding $p_i(\mathbf{x}) = 0, 1 \le i \le d$	Multivariate Polynomial Eigenproblem $P_i(\mathbf{x})\mathbf{v}_i = 0, 1 \le i \le d$

eigenproblems, univariate polynomial rootfinding, and matrix polynomial eigenproblems, have stable algorithms. Likewise, there are stable algorithms to solve linear systems of the form $A\mathbf{x} = \mathbf{b}$, which are multivariate.

One idea that works extremely well for univariate problems is to convert them into generalized eigenproblems. Instead of solving a rootfinding problem directly, one first constructs an eigenproblem whose eigenvalues match the roots. The companion matrix of a polynomial p is an example of this, as its characteristic polynomial is p, so its eigenvalues are the roots of p [21, Chapt. 7]. One can solve the companion eigenproblem using an eigensolver, which is one of the most reliable algorithms in scientific computing. For roots in $[-1, 1]$, this is a stable algorithm for univariate polynomial rootfinding [22,23].[1] Algebraic rootfinders attempt the same conversion, i.e., each algorithm converts eq. (1.1) into one or more generalized eigenvalue problems (GEPs). For polynomial systems in eq. (1.1) in d variables, one usually constructs d GEPs, the eigenvalues of which give the coordinates of each root. The Macaulay resultant method (see section 7) is an exception as it constructs a single GEP and extracts the roots from the eigenvectors, not eigenvalues. Analogously to the univariate case, one hopes that the eigenproblems are as well-conditioned as the original rootfinding problem. Unfortunately, this is not the case, which strongly suggests instability in the full algorithm.

Fig. 1 demonstrates that this suggested instability is consistently observed in practice for the most common bivariate algebraic rootfinders. Far from achieving any global stability, the most popular algorithms are not even stable for finding a single root at

[1] We focus on the companion matrix because it leads to a provably stable algorithm and provides a useful analogy to multivariate methods. There are many other univariate methods, such as [24,25], that may outperform the companion method.

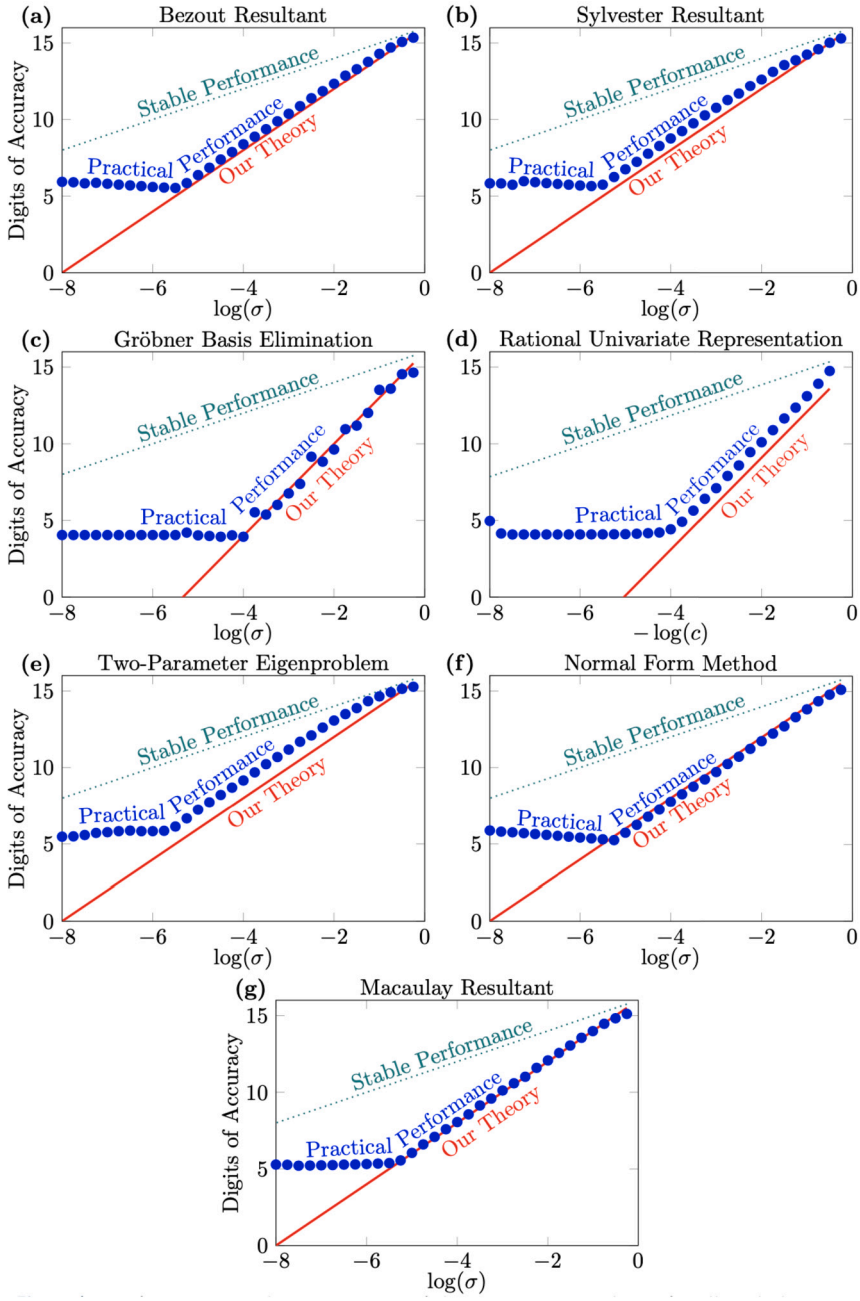


Fig. 1. Performance on a bivariate version of the system in Example 1.1 for all methods except for Gröbner basis elimination (Example 3.1) and the rational univariate representation (Example 4.1), with the root shifted to $(\frac{1}{3}, \frac{1}{3})$. The deviation of the practical performance from our theory is explained by the extreme proximity of the roots when σ is very small, which indicates that we should not expect conditioning analysis to be a good predictor. The methods are extremely inaccurate when σ is small.

$(\frac{1}{3}, \frac{1}{3})$, from simple systems generated by two quadratic polynomials.[2]

The first analysis of the instability of algebraic rootfinders appears in [1,2], which consider the Sylvester and Cayley/Bezout hidden-variable resultant methods. In particular, the authors study the following "devastating" example.

Example 1.1. Let Q be a $d \times d$ orthogonal matrix, $\sigma > 0$, and consider eq. (1.1) with

$$p_i(x_1, \ldots, x_d) = x_i^2 + \sigma \sum_{j=1}^{d} q_{ij} x_j, \quad 1 \le i \le d,$$

where q_{ij} is the (i, j) entry of Q. The system has a root at $(0, \ldots, 0)$.

By a conditioning analysis, one should expect to find the root at $(0, \ldots, 0)$ to within $\approx \sigma \mathbf{u}$, where $\mathbf{u}$ is the unit roundoff on a computer. However, it has been shown that Sylvester resultants can only achieve $\approx \sigma^{-2}\mathbf{u}$ when $d = 2$ and Cayley can only achieve $\approx \sigma^{-d}\mathbf{u}$ [2]. The eigenproblems constructed by these methods can be far more sensitive to perturbations than the original rootfinding problem, which is a hallmark of an unstable algorithm.[3,4]

Each algebraic rootfinder transforms eq. (1.1) into one or more GEPs, either directly or by way of a univariate rootfinding problem. For each method, we show that either a constructed eigenproblem or an intermediate univariate rootfinding problem can be ill-conditioned by a factor that depends exponentially on d. Our results are summarized in Table 2, where $\mathbf{x} = (x_1, \ldots, x_d)$ and $J(\mathbf{x}^*)$ denotes the Jacobian at the root $\mathbf{x}^*$; other notation is explained in the corresponding section, or for hidden-variable resultants in [1,2]. The last column of Table 2 is the ratio between a subproblem's condition number and that of the original root. By identifying a source of instability, we hope practitioners can focus on circumventing it.

2. Background

We begin with eigenproblems, conditioning, and algebraic geometry.

[2] For hidden-variable resultants, we implement the methods in [1,2]. For the two-parameter eigenproblem, we use the code from [26]. For normal form methods, we implement the null space method of [3–5]. We calculate the accuracy of solutions as $||\mathbf{x}^* - (\frac{1}{3}, \frac{1}{3})||_2$, with $\mathbf{x}^*$ the closest computed solution to $(\frac{1}{3}, \frac{1}{3})$. This only differs by a constant factor from the digits of accuracy of one solution coordinate, so it is reasonable to compare it to eigenvalue condition numbers. We generate orthogonal matrices randomly in MATLAB R2023a, setting the seed for the random number generator equal to 1. All the data is the median over 1000 trials for each value of σ or c.

[3] We note that for some choices of Q, such as $Q = I$, the problem splits trivially into univariate problems. Our analysis (and that of [2]) still holds for these systems, as the methods we consider do not take advantage of the potential splitting.

[4] The "devastating" example is designed to have a root that is close to a multiple root. While near-multiple roots are expected to cause numerical inaccuracy, as indicated by the condition number of the original root, the instability of these algorithms causes a far more devastating result than would be expected solely from the near-multiple root.

Table 2
The condition number of subproblems constructed by algebraic rootfinders.

Method	Condition Number of Subproblem	Devastating Example	Condition Number Ratio
Cayley Resultant (See [2])	$\kappa_{\text{eig}}(x_i^*) \geq \frac{\|\|V(\mathbf{x}^*)\|\|_2 \|\|W(\mathbf{x}^*)\|\|_2}{\|\det(J(\mathbf{x}^*))\|}$	$p(\mathbf{x}) = \mathbf{x}^2 + \sigma Q\mathbf{x}$ (See Example 1.1)	$\frac{\kappa_{\text{eig}}}{\kappa_{\text{root}}} \geq \sigma^{-d+1}$
Sylvester Resultant (2D) (See [2])	$\kappa_{\text{eig}}(x_i^*) \geq \frac{\|\|v(\mathbf{x}^*)\|\|_2 \|\|w(\mathbf{x}^*)\|\|_2}{\|\det(J(\mathbf{x}^*))\|}$	$p(\mathbf{x}) = \mathbf{x}^2 + \sigma Q\mathbf{x}$ (See Example 1.1)	$\frac{\kappa_{\text{eig}}}{\kappa_{\text{root}}} \geq \sigma^{-1}$
Gröbner Basis Elimination (See section 3)	(See section 3)	$p_i(\mathbf{x}) = x_i^2 - \sigma x_{i+1}^2$ (see Example 3.1)	$\frac{\kappa_{\text{uni}}}{\kappa_{\text{root}}} = \sigma^{-2^d+2}$
Rational Univariate (See section 4)	(See section 4)	$p(\mathbf{x}) = A\mathbf{x}^2 - \frac{A}{c^2 d}\mathbf{1}$ (see Example 4.1)	$\frac{\kappa_{\text{uni}}}{\kappa_{\text{root}}} \geq \frac{\|\|A\|\|_2}{\sqrt{d}} \left(\frac{c}{2}\right)^{2^d-2}$
Multiparameter Eigenvalue (See section 5)	$\kappa_{\text{eig}}(x_i^*) \geq \frac{\prod_{k=1}^{d} \prod_{j=1}^{n_k-1} \sigma_j^{(k)}(\mathbf{x}^*)}{\|\det(J(\mathbf{x}^*))\|}$	$p(\mathbf{x}) = \mathbf{x}^2 + \sigma P\mathbf{x}$ (see Example 5.1)	$\frac{\kappa_{\text{eig}}}{\kappa_{\text{root}}} \geq \sigma^{-d+1}$
Normal Form (See section 6)	$\kappa_{\text{eig}}(x_i^*) \geq \frac{\|\|[\det(Q)]_{\mathcal{B}}\|\|_2 \|\|\mathcal{B}(\mathbf{x}^*)\|\|_2}{\|\det(J(\mathbf{x}^*))\|}$	$p(\mathbf{x}) = \mathbf{x}^2 + \sigma P\mathbf{x}$ (see Example 6.1)	$\frac{\kappa_{\text{eig}}}{\kappa_{\text{root}}} \geq \sigma^{-d+1}$
Macaulay Resultant (See section 7)	$\kappa_{\text{eig}}(\lambda^*) \geq \frac{\|\|[\det(Q)]_{\mathcal{B}}\|\|_2 \|\|V(\mathbf{x}^*)\|\|_2}{\|\det(J(\mathbf{x}^*))h(\mathbf{x}^*)\|}$	$p(\mathbf{x}) = \mathbf{x}^2 + \sigma P\mathbf{x}$ (see Example 7.1)	$\frac{\kappa_{\text{eig}}}{\kappa_{\text{root}}} \geq \sigma^{-d+1}$

2.1. Generalized eigenproblems and matrix polynomials

Algebraic rootfinders convert eq. (1.1) into one or more GEPs. A GEP is given by $A\mathbf{x} = \lambda B\mathbf{x}$, where $A, B \in \mathbb{C}^{n\times n}$, and one seeks to find all the eigenvalue-eigenvector pairs $(\lambda, \mathbf{x})$ that satisfy $A\mathbf{x} = \lambda B\mathbf{x}$. When B is the identity matrix, a GEP simplifies to the standard eigenproblem $A\mathbf{x} = \lambda\mathbf{x}$.

Note that $A\mathbf{x} = \lambda B\mathbf{x}$ if and only if $(A - \lambda B)\mathbf{x} = 0$, so the eigenvalues are the values of λ for which $\det(A - \lambda B) = 0$. A nonzero vector $\mathbf{x}$ satisfying $(A - \lambda B)\mathbf{x} = 0$ is known as a right eigenvector, while a nonzero vector $\mathbf{y}$ satisfying $\mathbf{y}^\top(A - \lambda B) = 0$, where $\mathbf{y}^\top$ denotes the transpose of the vector $\mathbf{y}$, is known as a left eigenvector.

The form $A - \lambda B$ naturally generalizes in d variables to expressions of the form $V_0 + \sum_{i=1}^{d} x_i V_i$, where $V_i \in \mathbb{C}^{n\times n}$ for some integer n; these are known as linear matrix polynomials, and they form the basis for multiparameter eigenproblems (see section 5).

2.2. Condition numbers

For a function g, the normwise absolute condition number is a measurement of how much the output changes for small changes in input around x^*, i.e. [27, Chapt. 1],

$$\kappa(x^*) = \lim_{\epsilon\to 0} \sup_{||\delta x||_a < \epsilon} \frac{||g(x^* + \delta x) - g(x^*)||_b}{||\delta x||_a}, \tag{2.1}$$

where $||\cdot||_a$ and $||\cdot||_b$ denote norms on the input and output spaces. This definition also agrees with the definition of absolute asymptotic condition in [28]. The condition

number is unique once a norm is chosen for both the input and output, and tells you how much a solution can be perturbed given a perturbation of the input. Formally, a forward stable algorithm computes a solution with error at most $\approx \kappa(x^*)\mathbf{u}$, where $\mathbf{u}$ is the unit roundoff or machine epsilon. We will only consider forward stability, and will refer to algorithms as stable if they are forward stable.

We derive the absolute condition number of a root $\mathbf{x}^* = (x_1^*, \ldots, x_d^*)$ of eq. (1.1) as in [1]. Let $\mathbf{x}^* + \delta\mathbf{x}$ be a root of the perturbed polynomial system $p + \delta p$, where $\delta\mathbf{x} = [\delta x_1 \quad \cdots \quad \delta x_d]^\top$ and $\delta p = [\delta p_1 \quad \cdots \quad \delta p_d]^\top$. Then, we have up to first order

$$0 = (p+\delta p)(\mathbf{x}^*+\delta\mathbf{x}) = \begin{pmatrix} \frac{\partial p_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial p_1}{\partial x_d}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial p_d}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial p_d}{\partial x_d}(\mathbf{x}^*) \end{pmatrix} \begin{pmatrix} \delta x_1 \\ \vdots \\ \delta x_d \end{pmatrix} + \delta p(\mathbf{x}^*+\delta\mathbf{x}). \tag{2.2}$$

Now we choose the norm on the input polynomial system to be $||p|| = \max_{\mathbf{x}\in\mathbb{U}} ||p(\mathbf{x})||_2$, for a compact region $\mathbb{U} \subset \mathbb{C}^d$ containing a small ball around $\mathbf{x}^*$ that is fixed in advance, and the norm on the root (the output) to be the vector 2-norm. Then, applying the definition in eq. (2.1) gives

$$\kappa_{\text{root}}(\mathbf{x}^*) = \lim_{\epsilon\to 0} \sup_{||\delta_p||<\epsilon} \left\{ \frac{1}{\epsilon} \left|\left| \begin{pmatrix} \delta x_1 \\ \vdots \\ \delta x_d \end{pmatrix} \right|\right|_2 : (p+\delta p)(\mathbf{x}^*+\delta\mathbf{x}) = 0 \right\}. \tag{2.3}$$

Combining eq. (2.3) and eq. (2.2) gives

$$\kappa_{\text{root}}(\mathbf{x}^*) = ||(J(\mathbf{x}^*))^{-1}||_2, \quad J(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial p_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial p_1}{\partial x_d}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial p_d}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial p_d}{\partial x_d}(\mathbf{x}^*) \end{pmatrix}. \tag{2.4}$$

The matrix in eq. (2.2) and eq. (2.4) is the Jacobian matrix of eq. (1.1), which specializes in the univariate case to $\kappa_{\text{uni}}(x^*) = |(p_1'(x^*))^{-1}|$. For Gröbner basis elimination (see section 3) and the rational univariate representation (see section 4), we use this condition number to analyze the instability introduced by an intermediate univariate rootfinding problem.

The condition number we derive above is the same, given our choice of norms, as the condition number of [14,29,30]. Considering norms on the coefficient vector of p instead leads to closely related condition numbers; [31] gives a nice derivation in terms of the implicit function theorem. The condition number in eq. (2.4) is also used explicitly in [1,2,32] to analyze the instability of algebraic rootfinders. Beyond the established use in the literature, there is further justification for our choice. The analysis above demonstrates that the condition number does not depend on the choice of the region $\mathbb{U}$. In addition, [33, Cor. 2.3] implies that the limiting accuracy of Newton's method is directly proportional to $||(J(\mathbf{x}^*))^{-1}||_2\mathbf{u}$, which aligns with our practical observations.

The fact that local methods can achieve accuracy $||(J(\mathbf{x}^*))^{-1}||_2\mathbf{u}$ further justifies eq. (2.4) as a reasonable definition of the condition number for polynomial rootfinding; under our definitions, a stable algorithm can also be thought of as an algorithm that is as accurate as a local method.

Algebraic rootfinders convert eq. (1.1) into an eigenproblem. For multiparameter eigenproblems (see section 5), normal form methods (see section 6), and Macaulay resultants (see section 7), we are interested in the condition number of the constructed eigenproblem. We define the normwise absolute condition number for $A\mathbf{x} = \lambda B\mathbf{x}$ at eigenvalue λ^* as

$$\kappa_{\text{eig}}(\lambda^*) = \limsup_{\epsilon\to 0}\left\{\frac{||\Delta\lambda||_2}{\epsilon} : ((A+\Delta A) - (\lambda^*+\Delta\lambda)(B+\Delta B))\,(\mathbf{x}+\Delta\mathbf{x}) = 0, \right.$$
$$\left. ||\Delta A||_2, ||\Delta B||_2 \le \epsilon\right\}.$$

A similar definition is given in [2]. It is immediate from [34] that

$$\kappa_{\text{eig}}(\lambda^*) = \frac{||\mathbf{y}||_2||\mathbf{x}||_2}{|\mathbf{y}^\top B\mathbf{x}|}(1+|\lambda^*|), \tag{2.5}$$

where $\mathbf{y}$ and $\mathbf{x}$ are the left and right eigenvectors for the eigenvalue λ^*. In this formulation, the dependence of the condition number on A is encapsulated in the left and right eigenvectors.

A stable rootfinder should not construct any subproblem that is more ill-conditioned than the original rootfinding problem. The construction of an ill-conditioned subproblem strongly suggests that the algorithm is unstable, as the subproblem is more sensitive to perturbations than the original problem, though it does not rigorously prove instability. Unfortunately, we find that every algebraic rootfinder constructs either a univariate rootfinding problem or a GEP with a condition number worse than the original system's by a factor that depends exponentially on d; this suggests severe instability in the algorithms, which we observe in practice (see Fig. 1).

2.3. Zero-dimensional ideals and varieties

The polynomial system of d polynomials in d variables in eq. (1.1) generates a polynomial ideal $\langle p_1,\dots,p_d\rangle = \{f_1p_1+\cdots+f_dp_d : f_i \in \mathbb{C}[x_1,\dots,x_d]\}$. The system, or equivalently the ideal, is zero-dimensional if the variety $\mathcal{V}(p_1,\dots,p_d) = \{(x_1,\dots,x_d) \in \mathbb{C}^d : p_i(x_1,\dots,x_d) = 0, 1 \le i \le d\}$ consists of finitely many points. A root of a polynomial system is simple if the Jacobian $J(\mathbf{x}^*)$ is invertible. Suppose all the roots of eq. (1.1) are simple. In that case, being zero-dimensional is equivalent to the quotient $\mathbb{C}[x_1,\dots,x_d]/\langle p_1,\dots,p_d\rangle$ being a finite-dimensional vector space, whose dimension equals the number of roots. Throughout, our polynomial systems have only simple roots.

The equivalence in the last paragraph can be illustrated through the Lagrange interpolant basis for $\mathbb{C}[x_1,\dots,x_d]/\langle p_1,\dots,p_d\rangle$, which is important to later analysis. We call a polynomial $q_{\mathbf{x}}$ a Lagrange interpolant for the system at $\mathbf{x}$ if $q_{\mathbf{x}}(\mathbf{x}) \neq 0$ and $q_{\mathbf{x}}(\mathbf{x}') = 0$ for all $\mathbf{x}' \in \mathcal{V}(p_1,\dots,p_d)$, whenever $\mathbf{x}' \neq \mathbf{x}$. If a system is zero-dimensional with simple roots, then the Lagrange interpolants with $q_{\mathbf{x}}(\mathbf{x}) = 1$ are a basis for $\mathbb{C}[x_1,\dots,x_d]/\langle p_1,\dots,p_d\rangle$.

The radical of a polynomial ideal is denoted by $\sqrt{\langle p_1,\dots,p_d\rangle}$ and defined as $\sqrt{\langle p_1,\dots,p_d\rangle} = \{f \in \mathbb{C}[x_1,\dots,x_d] : f^m \in \langle p_1,\dots,p_d\rangle, m \in \mathbb{Z}_{\geq 0}\}$. An ideal is radical if $\langle p_1,\dots,p_d\rangle = \sqrt{\langle p_1,\dots,p_d\rangle}$. If $f^m \in \langle p_1,\dots,p_d\rangle$, then $f^m = f = 0$ in $\mathbb{C}[x_1,\dots,x_d]/\langle p_1,\dots,p_d\rangle$, so $f \in \langle p_1,\dots,p_d\rangle$. Thus, a zero-dimensional polynomial system with simple roots generates a radical ideal, so $\langle p_1,\dots,p_d\rangle$ is radical throughout.

In addition, we assume that the system in eq. (1.1) has no roots at infinity; this is generic, and any system with roots at infinity can be transformed via a random linear change of variables into one with all finite roots with probability one.[5]

3. Gröbner basis elimination

A Gröbner basis is a specific type of generating set for a polynomial ideal that allows one to deduce many important properties of the ideal. They are a popular technique for multivariate rootfinding [9, Chapt. 2]. For concreteness, consider solving eq. (1.1) using a Gröbner basis with respect to the lexicographical order. Since eq. (1.1) has a zero-dimensional variety, the Gröbner basis contains a univariate polynomial $g \in \langle p_1,\dots,p_d\rangle$ that generates the elimination ideal $\langle p_1,\dots,p_d\rangle \cap \mathbb{C}[x_i]$ and as such determines the x_i coordinates of the roots. However, this univariate polynomial can have extremely ill-conditioned roots, as demonstrated in the following example.

Example 3.1. Let $\sigma > 0$ and consider the system in eq. (1.1) with polynomials

$$p_i(x_1,\dots,x_d) = x_i^2 - \sigma x_{i+1}, \quad 1 \leq i \leq d-1, \quad p_d(x_1,\dots,x_d) = x_d^2 - \sigma x_1.$$

Expanding $p_{i-1}(x_1,\dots,x_d)\cdot(x_{i-1}^2 + \sigma x_i)$ shows that $x_i^2 = \sigma^{-2}x_{i-1}^4$ in the quotient space $\mathbb{C}[x_1,\dots,x_d]/\langle p_1,\dots,p_d\rangle$. Rearranging p_{i-1} also gives $x_i = \sigma^{-1}x_{i-1}^2$. Thus $x_i = \sigma^{-1}x_{i-1}^2 = \sigma^{-1}(\sigma^{-2}x_{i-2}^4) = \sigma^{-1}(\sigma^{-2}(\sigma^{-2}x_{i-3}^4)^2) = \cdots = \sigma^{-\sum_{k=0}^{d-1}2^k}x_i^{2^d} = \sigma^{-2^d+1}x_i^{2^d}$. Let $g = x_i^{2^d} - \sigma^{2^d-1}x_i$. We know that $g \in \langle p_1,\dots,p_d\rangle$ and has 2^d distinct roots, so g generates the elimination ideal $\langle p_1,\dots,p_d\rangle \cap \mathbb{C}[x_i]$. Due to the system's symmetry, the same univariate problem is generated for any ordering of the variables.

The root of the polynomial system at $(0,\dots,0)$ has absolute condition number given by $\kappa_{\text{root}}(0,\dots,0) = \sigma^{-1}$, but the root of g at 0 has absolute condition number

[5] The main reason for this assumption is the rootfinders are much simpler to present in the absence of roots at infinity. For example, for normal form and Macaulay resultant methods, additional care must be taken when roots are at infinity because the size of the constructed eigenproblem changes. All the rootfinders we consider can be adapted to work in the presence of roots at infinity, and we believe that much of our analysis extends to these variants, too.

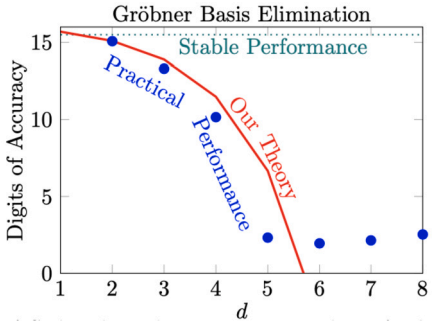


Fig. 2. Performance of Gröbner basis elimination on Example 3.1 for $d \geq 2$ and $\sigma = \frac{1}{2}$. We plot the practical performance against the theoretical performance of a stable algorithm and the prediction given by the analysis of Example 3.1. The deviation of the practical performance from our theory is explained by the extreme proximity of the roots for large values of d.

$\kappa_{\text{uni}}(0) = |g'(0)^{-1}| = \sigma^{-2^d+1}$. Thus, the Gröbner basis elimination method has generated a univariate rootfinding problem that is exponentially ill-conditioned relative to the original system, which suggests that the method is unstable. This instability is independent of the polynomial basis chosen and the order of the variables, i.e., it can not be avoided by representing g in a non-monomial basis or by changing the monomial order.

The instability is observed in two variables in Fig. 1 **(c)** for varying values of σ, and in Fig. 2 for $\sigma = \frac{1}{2}$ and varying values of d, confirming our theory.

4. Rational univariate representation

The rational univariate representation projects a multivariate rootfinding problem to a univariate one [8]. A projection is determined by a polynomial $t \in \mathbb{C}[x_1, \ldots, x_d]$ such that $t(\mathbf{x}) \neq t(\mathbf{x}')$ for distinct roots $\mathbf{x}$ and $\mathbf{x}'$ of eq. (1.1); we call this a separating polynomial. Given a separating polynomial, one finds the roots of $f(x) = \prod_{i=1}^{r}(x - t(\mathbf{x}_i))$, where r is the number of roots of the system. Unfortunately, we find that this univariate problem can be highly ill-conditioned.

In [8], it is proved that there always exists a linear separating polynomial. Thus, a separating polynomial has the form $t(x_1, \ldots, x_d) = u_1 x_1 + \cdots + u_d x_d$, where $u_i \in \mathbb{C}$. In addition, a scale assumption is necessary to obtain meaningful estimates for the absolute condition number. We make the particular choice $\sum_{i=1}^{d} |u_i|^2 \leq 1$.

4.1. A devastating example for the rational univariate representation

We focus on the following example for this section.

Example 4.1. For any invertible matrix $A \in \mathbb{C}^{d\times d}$ and $c > 0$, consider

$$p_i(x_1, \ldots, x_d) = \sum_{j=1}^{d} a_{ij}\left(x_j^2 - \frac{1}{c^2 d}\right), \quad 1 \le i \le d,$$

where a_{ij} is the (i,j) entry of A. The system has 2^d roots at $(\pm\frac{1}{c\sqrt{d}}, \ldots, \pm\frac{1}{c\sqrt{d}})$. The Jacobian at $\mathbf{x}^* = (\frac{1}{c\sqrt{d}}, \cdots, \frac{1}{c\sqrt{d}})$ is $J(\mathbf{x}^*) = \frac{2}{c\sqrt{d}}A$, so the absolute condition number of $\mathbf{x}^*$ is $\kappa_{\text{root}}(\mathbf{x}^*) = \frac{c\sqrt{d}}{2||A||_2}$. We prove that for any bounded linear projection of this system, the root $x^* = t(\mathbf{x}^*)$ corresponding to $\mathbf{x}^*$ has a condition number that is exponentially greater than $\kappa_{\text{root}}(\mathbf{x}^*)$.

Theorem 4.1. *Consider solving the system in Example 4.1 via the rational univariate representation using a linear projection $t(x_1, \ldots, x_d) = u_1x_1 + \cdots + u_dx_d$, with $\sum_{i=1}^{d}|u_i|^2 \le 1$, and let $f(x) = \prod_{i=1}^{r}(x - t(\mathbf{x}_i))$. Then, the root $x^* = t(\mathbf{x}^*)$ of f corresponding to $\mathbf{x}^*$ has condition number $\kappa_{uni}(x^*) \ge \left(\frac{c}{2}\right)^{2^d-1}$.*

Proof. Suppose we have a projection $t(x_1, \ldots, x_d) = u_1x_1 + \cdots + u_dx_d$ represented by a vector $\mathbf{u} = (u_1, \ldots, u_d)$. Let $[i \in S]$ be the indicator function that is 1 if $i \in S$ and 0 otherwise. The one-dimensional rootfinding problem is

$$f(x) = \prod_{S\subseteq[d]}\left(x - \frac{1}{c\sqrt{d}}\sum_{i=1}^{d}(-1)^{[i\in S]}u_i\right),$$

with $[d] = \{1, \ldots, d\}$. Differentiating $f(x)$ shows that the condition number of the root $x^* = \frac{1}{c\sqrt{d}}\sum_{i=1}^{d}u_i$ satisfies

$$\begin{aligned}
\kappa_{\text{uni}}(x^*)^{-1} &= \left|\prod_{\substack{S\subseteq[d]\\S\neq\emptyset}}\left(x^* - \frac{1}{c\sqrt{d}}\sum_{i=1}^{d}(-1)^{[i\in S]}u_i\right)\right| = \left|\prod_{\substack{S\subseteq[d]\\S\neq\emptyset}}\sum_{i\in S}\frac{2u_i}{c\sqrt{d}}\right| \\
&= \left(\frac{2}{c\sqrt{d}}\right)^{2^d-1}\left|\prod_{\substack{S\subseteq[d]\\S\neq\emptyset}}\sum_{i\in S}u_i\right| \le \left(\frac{2}{c\sqrt{d}}\right)^{2^d-1}\prod_{\substack{S\subseteq[d]\\S\neq\emptyset}}\sum_{i\in S}|u_i| \\
&\le \left(\frac{2}{c\sqrt{d}}\right)^{2^d-1}(||\mathbf{u}||_1)^{2^d-1} \le \left(\frac{2}{c\sqrt{d}}\right)^{2^d-1}\left(\sqrt{d}||\mathbf{u}||_2\right)^{2^d-1} \\
&\le \left(\frac{2}{c\sqrt{d}}\right)^{2^d-1}\left(\sqrt{d}\right)^{2^d-1} = \left(\frac{2}{c}\right)^{2^d-1}.
\end{aligned}$$

Thus $\kappa_{\text{uni}}(x^*) \ge \left(\frac{c}{2}\right)^{2^d-1}$. □

Thus, for the system in Example 4.1, the root $\mathbf{x}^* = (\frac{1}{c\sqrt{d}}, \cdots, \frac{1}{c\sqrt{d}})$ of the original system has $\kappa_{\text{root}}(\mathbf{x}^*) = \frac{c\sqrt{d}}{2||A||_2}$ but, if we try to solve this system using the rational univariate representation, the condition number of the corresponding root $x^* = t(\mathbf{x}^*)$ has $\kappa_{\text{uni}}(x^*) \geq \left(\frac{c}{2}\right)^{2^d-1}$. When $c > 2$, this suggests a loss in accuracy due to the univariate rootfinding problem that is exponential in d, which aligns with practical experiments when $d = 2$ (see Fig. 1 **(d)**). This instability is basis independent and can not be avoided by representing f in a different polynomial basis.

Choosing a nonlinear separating polynomial is also possible, which complicates the conditioning analysis. For each root $\mathbf{x}$, there exists a Lagrange interpolant $q_{\mathbf{x}}$ such that $q_{\mathbf{x}}(\mathbf{x}) = 1$ and $q_{\mathbf{x}}(\mathbf{x}') = 0$ for all roots $\mathbf{x}' \neq \mathbf{x}$. The degree of these interpolants can be bounded above (see section 6). Thus, if the degree of t is allowed to be large enough, then t can theoretically be chosen so that f is any univariate polynomial with r roots, so, in particular, the roots of f could be as well-conditioned as the original roots. However, we are unaware of a method for choosing such a clever separating polynomial. We suspect that finding such a separating polynomial is essentially as hard as finding the roots themselves, and as such, having a linear separating polynomial is a reasonable assumption. Still, the possibility of a more creative choice of separating polynomial is a potential way around this point of instability.

5. Multiparameter eigenproblem via operator determinants

Another method for solving a multidimensional rootfinding problem is to convert it into a multiparameter eigenproblem [7,35,36]. Given the polynomial system in eq. (1.1), we first construct a determinantal representation for each p_i, which is a linear matrix polynomial $W_i(x_1, \ldots, x_d)$ such that $\det(W_i) = p_i$. This gives a multiparameter eigenproblem of the form:

$$W_i(\mathbf{x})\mathbf{v}_i = \left(V_{i0} - \sum_{j=1}^{d} x_j V_{ij}\right)\mathbf{v}_i = 0, \quad 1 \leq i \leq d, \tag{5.1}$$

with $V_{ij} \in \mathbb{C}^{n_i \times n_i}$ for integers $n_1, \ldots, n_d$. A solution of the multiparameter eigenproblem consists of an eigenvalue $(x_1^*, \ldots, x_d^*)$ and eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_d$ that solve eq. (5.1). A d-tuple $(x_1^*, \ldots, x_d^*)$ is an eigenvalue of this multiparameter eigenproblem if and only if it is a root of the system in eq. (1.1). The usual approach to solve the multiparameter eigenproblem is via operator determinants [37]. One constructs

$$\Delta_0 = \begin{vmatrix} V_{11} & V_{12} & \cdots & V_{1d} \\ V_{21} & V_{22} & \cdots & V_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ V_{d1} & V_{d2} & \cdots & V_{dd} \end{vmatrix}, \quad \Delta_i = \begin{vmatrix} V_{11} & \cdots & V_{1,i-1} & V_{10} & V_{1,i+1} & \cdots & V_{1d} \\ V_{21} & \cdots & V_{2,i-1} & V_{20} & V_{2,i+1} & \cdots & V_{2d} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ V_{d1} & \cdots & V_{d,i-1} & V_{d0} & V_{d,i+1} & \cdots & V_{dd} \end{vmatrix},$$

where the vertical bars denote taking the block determinant with multiplication replaced by Kronecker products. Then, one solves the following GEPs:

$$(\Delta_i - x_i\Delta_0)\mathbf{z}_i = 0, \quad 1 \le i \le d. \tag{5.2}$$

5.1. Conditioning of the operator determinants method

Unfortunately, we show that the GEPs in eq. (5.2) can be exponentially ill-conditioned. Similarly to [38], we relate the conditioning of this problem to the matrix

$$B_0 = \begin{pmatrix} \mathbf{u}_1^\top V_{11}\mathbf{v}_1 & \mathbf{u}_1^\top V_{12}\mathbf{v}_1 & \cdots & \mathbf{u}_1^\top V_{1d}\mathbf{v}_1 \\ \mathbf{u}_2^\top V_{21}\mathbf{v}_2 & \mathbf{u}_2^\top V_{22}\mathbf{v}_2 & \cdots & \mathbf{u}_2^\top V_{2d}\mathbf{v}_2 \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{u}_d^\top V_{d1}\mathbf{v}_d & \mathbf{u}_d^\top V_{d2}\mathbf{v}_d & \cdots & \mathbf{u}_d^\top V_{dd}\mathbf{v}_d \end{pmatrix},$$

with $\mathbf{u}_i, \mathbf{v}_i$ the left and right eigenvectors associated with an eigenvalue $(x_1^*, \ldots, x_d^*)$. We assume the eigenvectors are all normalized so that $||\mathbf{u}_i||_2 = ||\mathbf{v}_i||_2 = 1$. We first prove the following extension of [38, Proposition 13], which relates B_0 to the Jacobian of eq. (1.1).

Proposition 5.1. *Let $W = (W_1, \ldots, W_d)$ be a multiparameter eigenproblem with simple eigenvalue $\mathbf{x}^* = (x_1^*, \ldots, x_d^*)$, where $W_i(\mathbf{x}) \in \mathbb{C}^{n_i \times n_i}$. Then*

$$B_0 = \begin{pmatrix} \pm\prod_{j=1}^{n_1-1}\sigma_j^{(1)}(\mathbf{x}^*) & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \pm\prod_{j=1}^{n_d-1}\sigma_j^{(d)}(\mathbf{x}^*) \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial p_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial p_1}{\partial x_d}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial p_d}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial p_d}{\partial x_d}(\mathbf{x}^*) \end{pmatrix}, \tag{5.3}$$

with $p_i = \det(W_i)$ and $\{\sigma_j^{(i)}\}_{j=1}^{n_i-1}$ the nonzero singular values of $W_i(\mathbf{x}^*)$.

Proof. The proof follows the argument in [38, Proposition 13]. Let $Z(t) = W_i(x_1^*, \cdots, x_{j-1}^*, t, x_{j+1}^*, \cdots, x_d^*)$ and note that $g(t) = \det(Z(t))$ is an analytic function of t, so we can calculate its derivative along any path. We choose $t \in x_j^* + \mathbb{I}$, for some small real interval $\mathbb{I}$. Then let $Z(t) = U(t)S(t)V(t)^H$, where U and V are unitary, $V(t)^H$ denotes the conjugate transpose of $V(t)$, S is a real diagonal matrix, and each term on the right-hand side is an analytic function of t for some (possibly smaller) interval around x_j^*. The decomposition can be permuted so that $S(x_j^*) = \mathrm{diag}\{\sigma_k(W_i(\mathbf{x}^*))\}_{k=1}^{n_i}$. It is a simple extension of [39, Theorem 1] that such a decomposition exists (see Appendix A), even if x_j^* is complex. This is in essence an analytic SVD, except that the diagonal entries of $S(t)$ may not be ordered.

Denote the last columns of $U(t)$, $V(t)$ by $\mathbf{u}_{n_i}(t)$, $\mathbf{v}_{n_i}(t)$, and the (n_i, n_i) entry of $S(t)$ by $\sigma_{n_i}(t)$. Then $\sigma_{n_i}(x_j^*) = 0$, $\mathbf{u}_{n_i}(x_j^*)^H = \mathbf{u}_i^\top$, $\mathbf{v}_{n_i}(x_j^*) = \mathbf{v}_i$, and $\sigma_{n_i}(t) = \mathbf{u}_{n_i}(t)^H Z(t)\mathbf{v}_{n_i}(t)$. In addition, $\mathbf{u}_{n_i}(x_j^*)Z(x_j^*) = Z(x_j^*)\mathbf{v}_{n_i}(x_j^*) = 0$, so

$$\frac{d\sigma_{n_i}}{dt}(x_j^*) = -\mathbf{u}_i^\top V_{ij}\mathbf{v}_i = -(B_0)_{ij}.$$

Then note that $g(t) = \det(Z(t)) = \mp\prod \sigma_i(t)$, so that

$$\frac{\partial p_i}{\partial x_j}(\mathbf{x}^*) = \frac{dg}{dt}(x_j^*) = \pm\prod_{k=1}^{n_i-1}\sigma_k^{(i)}(\mathbf{x}^*)(B_0)_{ij},$$

and the result follows. □

Note that without the assumption of a simple eigenvalue, the diagonal matrix in eq. (5.3) is not invertible, but we still have

$$\begin{pmatrix} \pm\prod_{j=1}^{n_1-1}\sigma_j^{(1)}(\mathbf{x}^*) & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \pm\prod_{j=1}^{n_d-1}\sigma_j^{(d)}(\mathbf{x}^*) \end{pmatrix} B_0 = \begin{pmatrix} \frac{\partial p_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial p_1}{\partial x_d}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial p_d}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial p_d}{\partial x_d}(\mathbf{x}^*) \end{pmatrix},$$

which shows that $\mathbf{x}^*$ is a simple root of eq. (1.1) only if $\mathbf{x}^*$ is a simple eigenvalue of the corresponding multiparameter eigenproblem. Thus, the system in eq. (1.1) always produces multiparameter eigenproblems that satisfy the conditions of Proposition 5.1, and in particular $W_i(\mathbf{x}^*)$ always has $n_i - 1$ nonzero singular values.

We can now characterize the conditioning of the GEPs in eq. (5.2).

Theorem 5.1. *Let $p = (p_1, \ldots, p_d)$ be a zero-dimensional polynomial system in eq. (1.1) with all simple roots and no roots at infinity. Let $W = (W_1, \ldots, W_d)$ be a multiparameter eigenproblem with $\det(W_k) = p_k$. Then the condition number of a GEP in eq. (5.2) generated by operator determinants is*

$$\kappa_{eig}(x_i^*) = |\det(B_0)^{-1}|(1 + |x_i^*|) = \frac{\prod_{k=1}^{d}\prod_{j=1}^{n_k-1}\sigma_j^{(k)}(\mathbf{x}^*)}{|\det(J(\mathbf{x}^*))|}(1 + |x_i^*|),$$

where J is the Jacobian of p and $\{\sigma_j^{(k)}\}_{j=1}^{n_k-1}$ are the $n_k - 1$ nonzero singular values of $W_k(\mathbf{x}^*)$.

Proof. The eigenvectors of eq. (5.2) are the Kronecker products of the eigenvectors of the original multiparameter eigenproblem [37], so the condition number of one of the resulting GEPs is

$$\kappa_{\text{eig}}(x_i^*) = \frac{||\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_d||_2||\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_d||_2}{|(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_d)^\top \Delta_0(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_d)|}(1 + |x_i^*|),$$

where $\mathbf{u}_i, \mathbf{v}_i$ are the normalized left and right eigenvectors corresponding to $\mathbf{x}^*$. For any matrices $V_1, \ldots, V_d$ with $V_i \in \mathbb{C}^{n_i \times n_i}$ a simple calculation gives

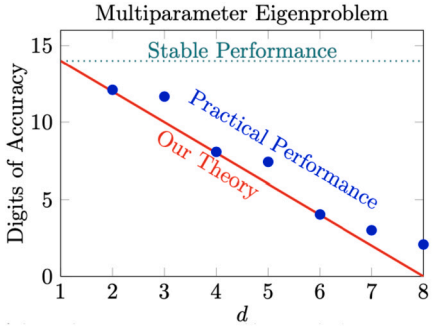


Fig. 3. Performance of the multiparameter eigenproblem method on Example 5.1 for $d \geq 2$ and $\sigma = \frac{1}{100}$. We plot the practical performance against the theoretical performance of a stable algorithm and the prediction given by Theorem 5.1.

$$(\mathbf{u}_1^\top V_1 \mathbf{v}_1) \cdots (\mathbf{u}_d^\top V_d \mathbf{v}_d) = (\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_d)^\top (V_1 \otimes \cdots \otimes V_d)(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_d).$$

Thus $(\mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_d)^\top \Delta_0 (\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_d) = \det(B_0)$, and the result follows from Proposition 5.1. □

Now, we consider an example that demonstrates how the formula for $\kappa_{\text{eig}}(x_i^*)$ in Theorem 5.1 leads to exponential instability.

Example 5.1. Let P be a $d \times d$ permutation matrix and $\sigma > 0$. For each i, there is an integer j_i such that the (i, j_i) entry of P equals 1. Consider the system in eq. (1.1) with

$$p_i(x_1, \ldots, x_d) = x_i^2 + \sigma x_{j_i}, \quad 1 \leq i \leq d.$$

The system has a root at $(0, \ldots, 0)$ with condition number $\kappa_{\text{root}}(0, \ldots, 0) = ||J(0, \ldots, 0)^{-1}||_2 = \sigma^{-1}$. We can construct a determinantal representation of this system with

$$W_i(x_1, \ldots, x_d) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} x_i + \begin{pmatrix} 0 & \sigma \\ 0 & 0 \end{pmatrix} x_{j_i} + \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, \quad 1 \leq i \leq d.$$

Theorem 5.1 proves that the condition number of the GEP that results from the operator determinants method is $\kappa_{\text{eig}}(0) \geq \sigma^{-d} \prod_{k=1}^{d} \prod_{j=1}^{n_k - 1} \sigma_j(V_{k0}) = \sigma^{-d}$. So, κ_{eig} for the GEP in eq. (5.2) can be greater than κ_{root} by a factor that depends exponentially on d. This suggested instability is observed in practical experiments (see Fig. 1 **(e)** and Fig. 3). As in section 3 and section 4, this instability is basis independent.

While Theorem 5.1 does not demonstrate that all determinantal representations of this system lead to exponential ill-conditioning, there is no known method that uses the factor $\prod_{k=1}^{d} \prod_{j=1}^{n_k - 1} \sigma_j^{(k)}(\mathbf{x}^*)$ to control the conditioning of the GEP in eq. (5.2). In fact, we believe that any such construction is likely as difficult as finding the roots themselves.

We have focused on the conditioning of the operator determinants method in this section rather than the conditioning of the original multiparameter eigenproblem, which by the analysis of [38] can be as well-conditioned as the original system, so our analysis does not rule out the possibility of a better method for solving multiparameter eigenproblems. However, the operator determinants method is the only global method for solving the multiparameter eigenproblem that we know. As an additional consequence, this observation suggests that there is no known stable global method to solve multiparameter eigenproblems.

6. Normal form methods

The structure of $\mathbb{C}[x_1,\dots,x_d]/\langle p_1,\dots,p_d\rangle$ gives rise to another class of rootfinding techniques, known as normal form methods [3–5]. Given a zero-dimensional polynomial system in eq. (1.1), $\mathbb{C}[x_1,\dots,x_d]/\langle p_1,\dots,p_d\rangle$ is a finite-dimensional $\mathbb{C}$-vector space with dimension equal to the number of roots of the system and the map given by multiplication by x_i is a linear map [3]. Given a basis $\mathcal{B}$ for $\mathbb{C}[x_1,\dots,x_d]/\langle p_1,\dots,p_d\rangle$, we can construct the matrix M_{x_i} for this map; these are known as multiplication or Möller–Stetter matrices [40,41].

We can construct a basis for $\mathbb{C}[x_1,\dots,x_d]/\langle p_1,\dots,p_d\rangle$ consisting of Lagrange interpolants. In this basis, the matrices M_{x_i} are diagonal with entries that are the coordinates of the roots. Therefore, the eigenvalues of the multiplication matrices in any basis $\mathcal{B}$ are the coordinates of the roots. A normal form method solves the system in eq. (1.1) by constructing, and then solving, the multiplication matrix eigenproblems

$$M_{x_i}\mathbf{w} = x_i\mathbf{w}, \quad 1 \le i \le d. \tag{6.1}$$

6.1. Conditioning of the multiplication matrix eigenproblems

To analyze the conditioning of the multiplication matrix eigenproblems in eq. (6.1) we need to characterize the left and right eigenvectors of the multiplication matrices. For a given basis $\mathcal{B}$, the left eigenvector corresponding to a coordinate x_i^* of a particular root $\mathbf{x}^* = (x_1^*,\dots,x_d^*)$ is $\mathcal{B}(\mathbf{x}^*)$. The right eigenvector is $[q_{\mathbf{x}^*}]_{\mathcal{B}}$, where $q_{\mathbf{x}^*}$ is the Lagrange interpolant for $p_1,\dots,p_d$ that vanishes at every root of $p_1,\dots,p_d$ except for $\mathbf{x}^*$, and $[q_{\mathbf{x}^*}]_{\mathcal{B}}$ denotes the expression of the interpolant in the basis $\mathcal{B}$, also known as its normal form. We begin with an extension of [32, Lemma 7.1] that gives the Lagrange interpolants. We use the notation $\mathbb{C}[x_1,\dots,x_d]^{d\times d}$ to denote the space of $d\times d$ matrices with entries in $\mathbb{C}[x_1,\dots,x_d]$.

Proposition 6.1. *Let $p = (p_1,\dots,p_d)$ be a zero-dimensional polynomial system in eq. (1.1) with all simple roots and no roots at infinity. Let $\mathbf{x}^* = (x_1^*,\dots,x_d^*)$ be a root and write p in the form*

$$\begin{pmatrix} p_1(x_1,\dots,x_d) \\ \vdots \\ p_d(x_1,\dots,x_d) \end{pmatrix} = Q \begin{pmatrix} x_1 - x_1^* \\ \vdots \\ x_d - x_d^* \end{pmatrix} \tag{6.2}$$

for some $Q \in \mathbb{C}[x_1,\dots,x_d]^{d\times d}$. *Then the Lagrange interpolant that vanishes at every root of* p *except* $(x_1^*,\dots,x_d^*)$ *is* $\det(Q)$.

Proof. By evaluating eq. (6.2), it is clear that $\det(Q)$ must vanish at any root of the system other than $(x_1^*,\dots,x_d^*)$ as the left hand side vanishes and the vector on the right hand side is nonzero. And $\det(Q)(x_1^*,\dots,x_d^*) = J(x_1^*,\dots,x_d^*) \neq 0$ because $(x_1^*,\dots,x_d^*)$ is a simple root, which proves the desired result as these properties uniquely define the Lagrange interpolant. □

This implies that the right eigenvector given above can be expressed as $[\det(Q)]_{\mathcal{B}}$. This form enables the following complete characterization of the conditioning of the multiplication matrix eigenproblems in eq. (6.1).

Theorem 6.1. *Let* $p = (p_1,\dots,p_d)$ *be a zero-dimensional polynomial system in eq.* (1.1) *with all simple roots and no roots at infinity. Let* $\mathbf{x}^* = (x_1^*,\dots,x_d^*)$ *be a root and write* p *in the form of eq.* (6.2). *Let* M_{x_i} *be a multiplication matrix for* p *with respect to a basis* $\mathcal{B}$. *Then, the absolute condition number of the eigenvalue* x_i^* *is*

$$\kappa_{eig}(x_i^*) = \frac{||[\det(Q)]_{\mathcal{B}}||_2 ||\mathcal{B}(\mathbf{x}^*)||_2}{|\det(J(\mathbf{x}^*))|}(1+|x_i^*|),$$

where J *is the Jacobian of* p.

Proof. Note that $[\det(Q)]_{\mathcal{B}}^{\top}\mathcal{B}(\mathbf{x}^*) = \det(Q(\mathbf{x}^*)) = \det(J(\mathbf{x}^*))$, so the result is a direct application of eq. (2.5) with $B = I$. □

We use this to demonstrate that normal form methods are exponentially unstable under a reasonable assumption on $\mathcal{B}$. In [3–5], a basis is selected from a set of basis monomials of degree $< \rho = \left(\sum_{i=1}^{d} \deg(p_i)\right) - d + 1$. We assume that the basis is selected from a pre-chosen set of monomials in some manner; [3–5] provide several clever numerical methods. Our analysis applies to any method that selects the basis from a pre-chosen set of this form, regardless of how the basis is subselected. For any reasonable set of basis monomials, any basis $\mathcal{B}$ selected in this manner additionally satisfies $||\mathcal{B}(0,\dots,0)||_2 \geq 1$.

Example 6.1. Let P be a $d \times d$ permutation matrix and $\sigma > 0$. For each i, there is an integer j_i such that the (i, j_i) entry of P equals 1. Consider the system in eq. (1.1) with

$$p_i(x_1,\dots,x_d) = x_i^2 + \sigma x_{j_i}, \quad 1 \leq i \leq d.$$

The system has a root at $(0,\dots,0)$ with condition number $\kappa_{\text{root}}(0,\dots,0) = ||J(0,\dots,0)^{-1}||_2 = \sigma^{-1}$. We show that the condition number of the resulting multiplication matrix eigenproblems in eq. (6.1) is greater by a factor that depends exponentially on d. For this system, $\rho = d+1$, so it is possible to reduce any polynomial of degree $\leq d-1$ to its normal form using multiples $f_k p_k$ with $f_k \in \mathbb{C}[x_1,\dots,x_d]$ and $\deg(f_k) \leq d-1$ [3]. Let $I = \langle p_1,\dots,p_d\rangle$. In particular, $\prod_{i=1}^d x_i - \left[\prod_{i=1}^d x_i\right]_{\mathcal{B}} \in I$ and is of degree $\leq d$ so

$$\prod_{i=1}^{d} x_i - \left[\prod_{i=1}^{d} x_i\right]_{\mathcal{B}} = \sum_{k=1}^{d} f_k p_k,$$

with $\deg(f_k) \leq d-1$, and where $[q]_{\mathcal{B}}$ denotes the expression of the projected polynomial $q + I \in \mathbb{C}[x_1,\dots,x_d]/I$ with respect to the basis $\mathcal{B}$. If the left-hand side does not vanish identically, then the monomial $\prod_{i=1}^d x_i$ must appear as a summand on the right-hand side. The only way that this can happen is if a term $\sigma^{-1}\left(\prod_{\substack{i=1\\ i\neq j_\ell}}^d x_i\right)(x_\ell^2 - \sigma x_{j_\ell})$ appears. Subtracting this term from both sides, we obtain

$$-\sigma^{-1}\left(\prod_{\substack{i=1\\ i\neq j_\ell}}^{d} x_i\right)(x_\ell^2) - \left[\prod_{i=1}^{d} x_i\right]_{\mathcal{B}} = \sum_{k=1}^{d} f_k' p_k.$$

Now if the left-hand side does not vanish identically, then the monomial $\sigma^{-1}\left(\prod_{\substack{i=1\\ i\neq j_\ell}}^d x_i\right)(x_\ell^2)$ must appear as a summand on the right-hand side. The only way that this can happen is if the same term $\sigma^{-1}\left(\prod_{\substack{i=1\\ i\neq j_\ell}}^d x_i\right)(x_\ell^2 - \sigma x_{j_\ell})$ appears on the right-hand side. In this way, the only possible reduction is to repeatedly substitute $\sigma^{-1}x_\ell^2$ for x_{j_ℓ} and then substitute back σx_{j_ℓ} for x_ℓ^2. As the degree of basis elements is $\leq d$, this implies that the normal form of $\prod_{i=1}^d x_i$ must be $\prod_{i=1}^d x_i$.

If $\prod_{i=1}^d x_i$ appears in the normal form of $\det(Q) - \prod_{i=1}^d x_i$, then it would be possible to reduce $\prod_{i=1}^d x_i$ to a monomial of lower degree using multiples $f_k p_k$ with $\deg(f_k) \leq d-1$, which it is not. Thus $\prod_{i=1}^d x_i$ appears in any normal form of $\det(Q)$, so $||[\det(Q)]_{\mathcal{B}}||_2 \geq 1$. By assumption, $||\mathcal{B}(0,\dots 0)||_2 \geq 1$, so $\kappa_{\text{eig}}(0) \geq \sigma^{-d}$, while $\kappa_{\text{root}}(0,\dots,0) = ||J(0,\dots,0)^{-1}||_2 = \sigma^{-1}$.

This suggests that the multiplication matrix eigenproblem can introduce instability that depends exponentially on d. We observe this instability in practical experiments (see Fig. 1 **(f)**).

One might hope to avoid the instability by selecting a clever basis. In particular, we know that the multiplication matrices in the Lagrange interpolant basis are diagonal, so the eigenproblem is trivial. However, finding the Lagrange interpolants is, in our estimation, equally as challenging as finding the roots. All known algorithms for computing

multiplication matrices preselect basis monomials from which to choose the basis $\mathcal{B}$ in the manner of [3–5]. Nevertheless, clever basis selection is a potential way around the instability. We address a few other interesting subtleties of normal form methods in section 8.

7. Macaulay resultant

The Macaulay resultant method (also called the u-resultant) forms an eigenproblem directly from the Macaulay matrix, defined in the monomial basis [6]. Consider a polynomial system in eq. (1.1). Let $h = \mu_0 + \sum_{i=1}^{d} \mu_i x_i$, and $\rho = \left(\sum_{i=1}^{d} \deg(p_i)\right) - d + 1$. For each p_i (and for h), M has rows consisting of the monomial coefficients of multiples of p_i (and h) by all monomials of degree $\leq \rho - \deg(p_i)$ ($\leq \rho - \deg(h)$). Consider the system in Example 1.1 in two variables, and let $h = \mu_0 + \mu_1 x + \mu_2 y$. Then, the Macaulay matrix in the monomial basis is

$$M = \begin{array}{c} \\ p_1 \\ xp_1 \\ yp_1 \\ p_2 \\ xp_2 \\ yp_2 \\ h \\ xh \\ yh \\ x^2h \\ xyh \\ y^2h \end{array} \begin{array}{c} \begin{array}{cccccccccc} 1 & x & y & x^2 & xy & y^2 & x^3 & x^2y & xy^2 & y^3 \end{array} \\ \begin{pmatrix} & \sigma q_{11} & \sigma q_{12} & 1 & & & & & & \\ & & & \sigma q_{11} & \sigma q_{12} & & 1 & & & \\ & & & & \sigma q_{11} & \sigma q_{12} & & 1 & & \\ & \sigma q_{21} & \sigma q_{22} & & & 1 & & & & \\ & & & \sigma q_{21} & \sigma q_{22} & & & & 1 & \\ & & & & \sigma q_{21} & \sigma q_{22} & & & & 1 \\ \mu_0 & \mu_1 & \mu_2 & & & & & & & \\ & \mu_0 & & \mu_1 & \mu_2 & & & & & \\ & & \mu_0 & & \mu_1 & \mu_2 & & & & \\ & & & \mu_0 & & & \mu_1 & \mu_2 & & \\ & & & & \mu_0 & & & \mu_1 & \mu_2 & \\ & & & & & \mu_0 & & & \mu_1 & \mu_2 \end{pmatrix} \end{array}.$$

The matrix M has more rows than columns. However, there exists a square submatrix of M that is singular if and only if $p_1, \ldots, p_d, h$ have a common root [6]. To obtain a square matrix, we drop some of the rows corresponding to h, which must be selected so that the resulting matrix is nonsingular [6]. When $d = 2$, the authors in [6] provide a careful method for selecting the rows to drop. Our analysis is valid for any dropped rows that give a nonsingular submatrix. Once we have a square matrix, we set $\mu_i = \alpha_i - \lambda\beta_i$ for α_i, β_i chosen randomly. For the example above, dropping the x^2h and y^2h rows gives

$$M(\lambda) = \begin{array}{c} \\ p_1 \\ xp_1 \\ yp_1 \\ p_2 \\ xp_2 \\ yp_2 \\ h \\ xh \\ yh \\ xyh \end{array} \begin{array}{c} \begin{array}{cccccccccc} 1 & x & y & x^2 & xy & y^2 & x^3 & x^2y & xy^2 & y^3 \end{array} \\ \begin{pmatrix} & \sigma q_{11} & \sigma q_{12} & 1 & & & & & & \\ & & & \sigma q_{11} & \sigma q_{12} & & 1 & & & \\ & & & & \sigma q_{11} & \sigma q_{12} & & 1 & & \\ & \sigma q_{21} & \sigma q_{22} & & & 1 & & & & \\ & & & \sigma q_{21} & \sigma q_{22} & & & & 1 & \\ & & & & \sigma q_{21} & \sigma q_{22} & & & & 1 \\ \alpha_0 - \lambda\beta_0 & \alpha_1 - \lambda\beta_1 & \alpha_2 - \lambda\beta_2 & & & & & & & \\ & \alpha_0 - \lambda\beta_0 & & \alpha_1 - \lambda\beta_1 & \alpha_2 - \lambda\beta_2 & & & & & \\ & & \alpha_0 - \lambda\beta_0 & & \alpha_1 - \lambda\beta_1 & \alpha_2 - \lambda\beta_2 & & & & \\ & & & & \alpha_0 - \lambda\beta_0 & & & \alpha_1 - \lambda\beta_1 & \alpha_2 - \lambda\beta_2 & \end{pmatrix} \end{array}.$$

This is a GEP

$$M(\lambda)\mathbf{x} = (A - \lambda B)\mathbf{x} = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} - \lambda \begin{bmatrix} 0 \\ B_2 \end{bmatrix} = 0. \tag{7.1}$$

This GEP only has rank(B) standard eigenvalues; we call these finite eigenvalues. For the Macaulay eigenproblem, the right eigenvectors corresponding to finite eigenvalues are the column labels evaluated at a root $\mathbf{x}^*$. We examine the conditioning of this eigenproblem.

7.1. Conditioning of the Macaulay resultant eigenproblem

Although the roots are extracted from the eigenvectors, we have found it easier to examine eigenvalue conditioning. While the eigenvalue condition number may not bound the eigenvector condition number, with a mild assumption, the eigenvalue condition number bounds the condition number of finding the root from this method, as follows. Suppose we know the coordinates of a root $\mathbf{x}^* = (x_1^*, \ldots, x_d^*)$. The eigenvalue λ^* corresponding to this root is the solution to $h(\lambda)(\mathbf{x}^*) = 0$. We assume all the randomly chosen coefficients α_i, β_i are $O(1)$, so solving this simple linear equation for λ^* is well-conditioned. Thus, the eigenvalue condition number provides a lower bound on the condition number of finding the root $\mathbf{x}^*$ from the GEP in eq. (7.1).

As stated above, the right eigenvector corresponding to a root $\mathbf{x}^*$ is the evaluation of the column labels at the root $\mathbf{x}^*$. An eigenvalue is a choice of λ for which $h(\lambda^*)$ has a common root with $p_1, \ldots, p_d$. Then, the left eigenvector expresses a combination

$$\sum_{i=1}^{d} P_i p_i + Hh(\lambda^*) = 0, \tag{7.2}$$

where $P_i, H \in \mathbb{C}[x_1, \ldots, x_d]$, so that in particular $Hh(\lambda^*) \in \langle p_1, \ldots, p_d \rangle$. Thus, $Hh(\lambda^*)$ must vanish at all roots of the system. However, $h(\lambda^*)$ only vanishes at $\mathbf{x}^*$, so H vanishes at all roots $\neq \mathbf{x}^*$. Recall that, by our assumption that all roots are simple, $\langle p_1, \ldots, p_d \rangle$ is radical. Thus, either $H \in \langle p_1, \ldots, p_d \rangle$ or H is the Lagrange interpolant for the system at $\mathbf{x}^*$. If $H \in \langle p_1, \ldots, p_d \rangle$, then for all λ there exists a combination as in eq. (7.2), which implies that the eigenproblem is singular. Thus, H is the Lagrange interpolant, given by the same formula as in section 6. That is, if we write the system as in eq. (6.2) for some polynomial matrix Q, then $H = \det(Q)$ in $\mathbb{C}[x_1, \ldots, x_d]/\langle p_1, \cdots, p_d \rangle$.

We want to understand how the equivalence in $\mathbb{C}[x_1, \ldots, x_d]/\langle p_1, \cdots, p_d \rangle$ plays out. For this, we note that the number of finite eigenvalues of the problem $A - \lambda B$ must equal the number of roots of eq. (1.1). This implies that the number of rows corresponding to h is equal to the dimension of $\mathbb{C}[x_1, \ldots, x_d]/\langle p_1, \ldots, p_d \rangle$. The monomials corresponding to these rows must be independent in $\mathbb{C}[x_1, \ldots, x_d]/\langle p_1, \cdots, p_d \rangle$ to have a nonsingular eigenproblem. Hence, they are a basis $\mathcal{B}$ for $\mathbb{C}[x_1, \ldots, x_d]/\langle p_1, \ldots, p_d \rangle$. Then $H = [\det(Q)]_{\mathcal{B}}$. We conclude the following, which is analogous to Theorem 6.1.

Theorem 7.1. *Let $p = (p_1, \dots, p_d)$ be a zero-dimensional polynomial system in eq. (1.1) with all simple roots and no roots at infinity. Let $\mathbf{x}^* = (x_1^*, \dots, x_d^*)$ be a root and write p in the form of eq. (6.2). Let $M = A - \lambda B$ be a Macaulay resultant eigenproblem in eq. (7.1) for p constructed with a random linear polynomial h and $\mathcal{B}$ the basis given by the rows corresponding to h. Then, the absolute condition number of the eigenvalue λ^* corresponding to the root $\mathbf{x}^* = (x_1^*, \dots, x_d^*)$ satisfies*

$$\kappa_{eig}(\lambda^*) \geq \frac{||[\det(Q)]_{\mathcal{B}}||_2 ||V(\mathbf{x}^*)||_2}{|\det(J(\mathbf{x}^*))h(\mathbf{x}^*)|},$$

where J is the Jacobian of p and $V(\mathbf{x}^*)$ is the evaluation of the column labels at $\mathbf{x}^*$.

Proof. From eq. (2.5) we have

$$\kappa_{\text{eig}}(\lambda^*) \geq \frac{||\mathbf{v}||_2 ||\mathbf{w}||_2}{|\mathbf{v}^\top B \mathbf{w}|},$$

where $\mathbf{v}$ and $\mathbf{w}$ are the left and right eigenvectors. Let the system have r roots. Because of the block structure of B, only the last r entries of $\mathbf{v}$ affect $\mathbf{v}^\top B \mathbf{w}$, which becomes $[\det(Q)]_{\mathcal{B}}^\top B_2 V(\mathbf{x}^*)$, where B_2 is the lower nonzero block of B consisting of the rows corresponding to h. Then $[\det(Q)]_{\mathcal{B}}^\top B_2$ is the coefficient vector of $[\det(Q)]_{\mathcal{B}} \cdot h$ with respect to the column labels, so $[\det(Q)]_{\mathcal{B}}^\top B_2 V(\mathbf{x}^*) = (\det(Q) \cdot h)(\mathbf{x}^*) = \det(J(\mathbf{x}^*))h(\mathbf{x}^*)$, and the result follows, noting $||v||_2 \geq ||[\det(Q)]_{\mathcal{B}}||_2$. □

We complete this section with a devastating example for the Macaulay resultant method, which is the same system as in Example 6.1. We assume that $||V(0, \dots, 0)||_2 \geq 1$, which holds for any reasonable polynomial basis.

Example 7.1. Let P be a $d \times d$ permutation matrix and $\sigma > 0$. For each i, there is an integer j_i such that the (i, j_i) entry of P equals 1. Consider the system in eq. (1.1) with

$$p_i(x_1, \dots, x_d) = x_i^2 + \sigma x_{j_i} \quad 1 \leq i \leq d.$$

The system has a root at $(0, \dots, 0)$ with condition number $\kappa_{\text{root}}(0, \dots, 0) = ||J(0, \dots, 0)^{-1}||_2 = \sigma^{-1}$. By the analysis in Example 6.1, $||[\det(Q)]_{\mathcal{B}}||_2 \geq 1$, and, by assumption, $||V(0, \dots, 0)||_2 \geq 1$. So, if λ^* is the eigenvalue corresponding to $(0, \dots, 0)$, then $\kappa_{\text{eig}}(\lambda^*) \geq \sigma^{-d} |h(0, \dots, 0)^{-1}|$.

Thus, there is an instability with a factor that depends exponentially on d, which is observed in our experiments (see Fig. 1 **(g)**). We refine some of these results in section 8.

8. Bivariate refinement of conditioning formulas for normal form methods and Macaulay resultant matrices

Examine the numerator of the formulas in Theorem 6.1 and Theorem 7.1. For normal form methods, if $||[\det(Q)]_{\mathcal{B}}||_2||\mathcal{B}(\mathbf{x}^*)||_2 \approx 1$, then the condition number is $\kappa_{\text{eig}}(x_i^*) \approx |\det(J(\mathbf{x}^*))^{-1}|$, which aligns with results from [2] and section 5. For the Macaulay resultant method, if $||[\det(Q)]_{\mathcal{B}}||_2||V(\mathbf{x}^*)||_2 \approx 1$ then $\kappa_{\text{eig}}(\lambda^*) \approx |\det(J(\mathbf{x}^*))^{-1}|$. However, this does not always occur. As discussed in section 6, it is in theory possible to select $\mathcal{B}$ to make the multiplication matrix eigenproblems trivial; the same is true for the Macaulay resultant eigenproblem. However, all known normal form algorithms choose basis monomials from which to subselect the basis $\mathcal{B}$, and similarly all known Macaulay resultant algorithms preselect the basis monomials for the Macaulay matrix, which typically makes $||\mathcal{B}(\mathbf{x}^*)||_2 = O(1)$ and $||V(\mathbf{x}^*)||_2 = O(1)$. Of more interest is the fact that $\det(Q)$ can reduce in unexpected ways in the basis $\mathcal{B}$, as demonstrated in the following example.

Example 8.1. Let A be a 2×2 orthogonal matrix and $\sigma > 0$. Consider

$$\begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \begin{pmatrix} x^2 + \sigma a_{11}x + \sigma a_{12}y \\ xy + \sigma y^2 + \sigma a_{21}x + \sigma a_{22}y \end{pmatrix} = \begin{pmatrix} x + \sigma a_{11} & \sigma a_{12} \\ y + \sigma a_{21} & \sigma y + \sigma a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Suppose that $\mathcal{B} = \{1, x, y, y^2\}$. Then $[\det(Q)]_{\mathcal{B}} = -\sigma^2 y^2 + \sigma(a_{22}x - a_{12}y) - \sigma^2 a_{21}x + \sigma^2(a_{11} - a_{22})y + \sigma^2$, so $||[\det(Q)]_{\mathcal{B}}||_2 = O(\sigma)$, and the resulting eigenvalue condition number for one of the multiplication matrix eigenproblems or for the Macaulay resultant eigenproblem is $O(\sigma^{-1})$ while $|\det(J(0,0))^{-1}| = \sigma^{-2}$.

Thus, the conditioning of the eigenproblems can meaningfully differ from $|\det(J(\mathbf{x}^*))^{-1}|$. In two dimensions, we prove that when the eigenvalue conditioning differs from $|\det(J(\mathbf{x}^*))^{-1}|$, other parts of the algorithms become unstable.

To facilitate this analysis, let $\hat{M}_\rho$ be the submatrix of the Macaulay matrix constructed as in section 7 consisting of only rows corresponding to multiples of $p_1, \ldots, p_d$. For the system in Example 1.1 in two variables:

$$\hat{M}_\rho = \begin{array}{c} \\ p_1 \\ xp_1 \\ yp_1 \\ p_2 \\ xp_2 \\ yp_2 \end{array} \begin{array}{c} \begin{array}{cccccccccc} 1 & x & y & x^2 & xy & y^2 & x^3 & x^2y & xy^2 & y^3 \end{array} \\ \begin{pmatrix} \sigma q_{11} & \sigma q_{12} & 1 & & & & & & & \\ & & \sigma q_{11} & \sigma q_{12} & & & 1 & & & \\ & & & & \sigma q_{11} & \sigma q_{12} & & 1 & & \\ \sigma q_{21} & \sigma q_{22} & & & & 1 & & & & \\ & & \sigma q_{21} & \sigma q_{22} & & & & & 1 & \\ & & & & \sigma q_{21} & \sigma q_{22} & & & & 1 \end{pmatrix} \end{array}.$$

Analogously, $\hat{M}_{\rho-1}$ consists of rows that are multiples of p_i by monomials of degree $\leq \rho - 1 - \deg(p_i)$. The next lemma helps connect the behavior of the system in Example 8.1 to $\hat{M}_\rho$.

Lemma 8.1. *Suppose that $I = \langle p_1, p_2 \rangle$ is a zero-dimensional radical ideal with no roots at infinity. Let $\rho = \deg(p_1) + \deg(p_2) - 1$. Then, the rows of the Macaulay matrix $\hat{M}_{\rho-1}$ span $I_{\leq \rho-1}$, the space of polynomials in I of degree $\leq \rho - 1$.*

Proof. We know from [3] and [42, Chapt. 1, p.46] that any polynomial in $I_{\leq \rho-1}$ can be generated by a linear combination of the rows of $\hat{M}_\rho$ because the rows of $\hat{M}_\rho$ span $I_{\leq \rho}$, so it suffices to prove that such a combination always generates a polynomial of degree exactly ρ if it includes rows that are in $\hat{M}_\rho \backslash \hat{M}_{\rho-1}$. Let $\deg(ap_1 + bp_2) < \rho$, where $a, b \in \mathbb{C}[x_1, x_2]$, a has degree $\rho - \deg(p_1)$, and b has degree $\rho - \deg(p_2)$. The degree ρ part is $a'p_1' + b'p_2' = 0$, where a', p_1', b', p_2' are the highest degree homogeneous terms of the respective polynomials. Because the ideal $\langle p_1, p_2 \rangle$ has no roots at infinity, we have that $\mathcal{V}(p_1', p_2') = \emptyset$. Thus p_1' and p_2' are relatively prime, so $p_1' \mid b'$ and $p_2' \mid a'$, which contradicts the degree assumptions. Thus if $ap_1 + bp_2$ has degree $< \rho$, then $\deg(a) < \rho - \deg(p_1)$ and $\deg(b) < \rho - \deg(p_2)$. □

With this, we can explain the system's behavior in Example 8.1. Recall that for the normal form and Macaulay resultant methods, a basis $\mathcal{B}$ for $\mathbb{C}[x_1, \ldots, x_d]/\langle p_1, \ldots, p_d \rangle$ is selected from a predetermined set of basis monomials. We have the following connection.

Proposition 8.1. *In two dimensions, with the setup as in Theorem 6.1 or Theorem 7.1, and assuming that p_1 and p_2 are scaled so that the maximum absolute value of their coefficients is at least 1, we have $||[\det(Q)]_{\mathcal{B}}||_2 \geq \sigma_{\min}(\hat{M}_\rho)$, with $\hat{M}_\rho$ the Macaulay matrix with respect to a set of basis monomials that contains $\mathcal{B}$.*

Proof. We can shift any root to $(0, 0)$ by setting $x_i' = x_i - x_i^*$, so it suffices to prove the proposition with $(x_1^*, x_2^*) = (0, 0)$. Note that

$$Adj(Q)Q = \begin{pmatrix} q_{22} & -q_{12} \\ -q_{21} & q_{11} \end{pmatrix} \begin{pmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \end{pmatrix} = \begin{pmatrix} \det(Q) & 0 \\ 0 & \det(Q) \end{pmatrix},$$

so $q_{22}f - q_{12}g = x_1 \det(Q)$ and $-q_{21}f + q_{11}g = x_2 \det(Q)$. These combinations are expressed as linear combinations of the rows of the Macaulay matrix $\hat{M}_\rho$.

As a consequence of Lemma 8.1, because $\deg([\det(Q)]_{\mathcal{B}}) \leq \rho - 1$ and $\deg(\det(Q)) \leq \rho - 1$, we can reduce $\det(Q)$ to $[\det(Q)]_B$ using the rows of the Macaulay matrix $\hat{M}_{\rho-1}$, which are combinations $x_1^{\alpha_1} x_2^{\alpha_2} p_i$ with $\alpha_1 + \alpha_2 \leq \rho - \deg(p_i) - 1$. We want to show that there is a choice of Q for which $\det(Q) = [\det(Q)]_{\mathcal{B}}$. It suffices to show for any α_1, α_2 that there exists Q' such that $Q' \binom{x_1}{x_2} = \binom{p_1}{p_2}$ and $\det(Q') = \det(Q) + x_1^{\alpha_1} x_2^{\alpha_2} p_1$. Note that $p_2 = p_2 + x_1^{\alpha_1+1} x_2^{\alpha_2+1} - x_1^{\alpha_1+1} x_2^{\alpha_2+1}$. We split these new terms, giving

$$Q' = \begin{pmatrix} q_{11} & q_{12} \\ q_{21} - x_2 x_1^{\alpha_1} x_2^{\alpha_2} & q_{22} + x_1 x_1^{\alpha_1} x_2^{\alpha_2} \end{pmatrix},$$

with $\det(Q') = \det(Q) + x_1^{\alpha_1} x_2^{\alpha_2} p_1$. In addition, because $\alpha_1 + \alpha_2 \leq \rho - \deg(p_1) - 1 = \deg(p_2) - 2$, this does not change the degree of p_2. Therefore, we can choose Q such

that $\det(Q)$ is reduced. This, with the scale assumption on the polynomials, proves that one of the coefficient vectors of the combinations $q_{22}p_1 - q_{12}p_2 = x_1\det(Q)$ and $-q_{21}p_1 + q_{11}p_2 = x_2\det(Q)$ in the Macaulay matrix $\hat{M}_\rho$ has magnitude at least 1, so $||[\det(Q)]_{\mathcal{B}}||_2 \geq \sigma_{\min}(\hat{M}_\rho)$. □

Now we have the following refinements of Theorem 6.1 and Theorem 7.1 for $d = 2$.

Theorem 8.1. *Let $p = (p_1, p_2)$ be a zero-dimensional polynomial system in eq. (1.1) with all simple roots and no roots at infinity. Let $\mathbf{x}^* = (x_1^*, x_2^*)$ be a root and write p in the form of eq. (6.2). Assume that p_1 and p_2 are scaled so that the maximum absolute value of their coefficients is at least 1. Let $M_{x_i} : \mathbb{C}[x_1, x_2]/\langle p\rangle \to \mathbb{C}[x_1, x_2]/\langle p\rangle$ be the multiplication matrix for multiplication by x_i in a basis $\mathcal{B}$. Then, the absolute condition number of the eigenvalue x_i^* satisfies*

$$\kappa_{eig}(x_i^*) \geq \frac{\sigma_{\min}(\hat{M}_\rho)||\mathcal{B}(x_1^*, x_2^*)||_2}{|\det(J(x_1^*, x_2^*))|},$$

where J is the Jacobian of p and $\hat{M}_\rho$ is the Macaulay matrix that is used to select $\mathcal{B}$.

Proof. It follows by substituting Proposition 8.1 into Theorem 6.1. □

Theorem 8.2. *Let $p = (p_1, p_2)$ be a zero-dimensional polynomial system in eq. (1.1) with all simple roots and no roots at infinity. Let $\mathbf{x}^* = (x_1^*, x_2^*)$ be a root and write p in the form of eq. (6.2). Assume that p_1 and p_2 are scaled so that the maximum absolute value of their coefficients is at least 1. Let $M = A - \lambda B$ be the Macaulay resultant eigenproblem in eq. (7.1) for p constructed with a random linear polynomial h. Then the absolute condition number of the eigenvalue λ^* corresponding to the root (x_1^*, x_2^*) satisfies*

$$\kappa_{eig}(\lambda^*) \geq \frac{\sigma_{\min}(A_1)||V(x_1^*, x_2^*)||_2}{|\det(J(x_1^*, x_2^*))h(x_1^*, x_2^*)|},$$

where J is the Jacobian of p and $V(x_1^*, x_2^*)$ is the evaluation of the column labels at (x_1^*, x_2^*).

Proof. It follows by substituting Proposition 8.1 into Theorem 7.1, noting that $\hat{M}_\rho = A_1$. □

Small singular values of $\hat{M}_\rho$ lead to conditioning issues for the two principal numerical constructions of multiplication matrices in [3–5]. These methods divide into column space methods, where one selects columns from $\hat{M}_\rho$ to make up an invertible submatrix, and null space methods, where one computes the null space of $\hat{M}_\rho$. Small singular values of $\hat{M}_\rho$ cause conditioning issues for column space methods, as any submatrix is near-singular. It also affects null space methods. It is proved in [14, Chapt. 14.3.2] that the

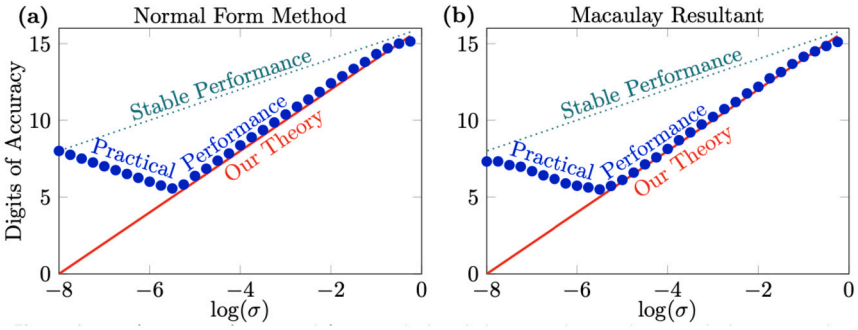


Fig. 4. Performance of a normal form method and the Macaulay resultant method on Example 8.1. We plot the practical performance against the theoretical performance of a stable algorithm. The deviation of observations from our prediction is because of the proximity of roots for small values of σ.

condition number of computing the null space of a matrix is directly related to the inverse of the smallest nonzero singular value. Thus, in two variables, by considering both the construction and the eigenproblem, the conditioning of normal form methods is still closely related to $|\det(J(\mathbf{x}))^{-1}|$. Practical experiments confirm this (see Fig. 4 **(a)**).

In [6], it is noted that the Macaulay eigenproblem can be reduced to eliminate the infinite eigenvalues. Given the form of eq. (7.1), we can find Z such that $A_1 Z = 0$; then the problem reduces to $(A_2 Z - \lambda B_2 Z)x = 0$. The problem of finding the eigenvalues for the original and reduced problem are equivalent. Moreover, the eigenvectors corresponding to finite eigenvalues give a basis for the right null space of A_1. As we are focused on global rootfinding, finding the eigenvectors for the original problem is equivalent to reducing (finding Z) and then solving the reduced problem. The reduced problem has the same eigenvalue condition number as the original, and computing Z has a condition number proportional to $(\sigma_{\min}(A_1))^{-1}$ (see [14, Chapt. 14.3.2]). We conclude that in two dimensions, while the condition number of the eigenproblem may differ from $|\det(J(\mathbf{x}^*))^{-1}|$, the conditioning of the eigenvectors, from which the roots are obtained, in general, does not. We observe this behavior in practice (see Fig. 4 **(b)**).

In higher dimensions, no characterization is possible in the manner of Theorem 8.1 and Theorem 8.2. The following example in three dimensions can easily be generalized and demonstrates unavoidable deviation of the conditioning of the eigenproblems from $|\det(J(\mathbf{x}^*))^{-1}|$.

Example 8.2. Let

$$\begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} = \begin{pmatrix} xy + \sigma x^2 + \sigma y \\ xy + \sigma y^2 + \sigma z \\ xy + \sigma z^2 + \sigma x \end{pmatrix} = \begin{pmatrix} y + \sigma x & \sigma & 0 \\ y & \sigma y & \sigma \\ y + \sigma & 0 & \sigma z \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

In the basis $\mathcal{B} = \{1, x, y, z, yz, xz, y^2 z, xyz\}$, the Lagrange interpolant is $[\det(Q)]_{\mathcal{B}} = \sigma^3 xyz + \sigma^2 y^2 z + \sigma^2 y - \sigma^2 yz + \sigma^3$. Then $||[\det(Q)]_{\mathcal{B}}||_2 = O(\sigma^2)$, so, for normal form methods, $\kappa_{\text{eig}}(0) = O(\sigma^{-1})$. However, in contrast to two dimensions, the smallest nonzero

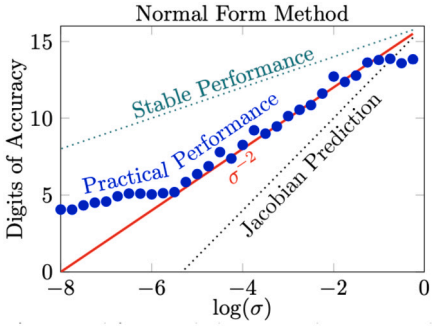


Fig. 5. Performance of a normal form method on Example 8.2. We plot the practical performance against the theoretical performance of a stable algorithm, the predicted performance given by the line σ^{-2}, and the prediction given by the Jacobian, which demonstrates that this example can be solved more accurately than would be predicted by a direct analogy of Theorem 8.1.

singular value of the Macaulay matrix $\hat{M}_\rho$ is only $O(\sigma^{-1})$. We can solve this in practice with an accuracy around σ^{-2} (see Fig. 5), which beats the estimate given by $\det(J(0,0,0)^{-1}) = \sigma^{-3}$. Thus, a high-dimensional refinement of Theorem 6.1 in the manner of Theorem 8.1 is impossible.

Declaration of competing interest

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Appendix A. An analytic decomposition for complex square matrices

We use the following extension of [39, Theorem 1] in the proof of Proposition 5.1 to relate the conditioning of the GEPs in eq. (5.2) to the Jacobian of the system in eq. (1.1).

Theorem A.1. *Let $Z(t)$ be a function of a complex variable t such that $Z(t) \in \mathbb{C}^{d\times d}$ and $Z(t)$ is analytic for all t in some real interval around a point t_0. Then there exists a decomposition on some (possibly distinct) real interval around t_0*

$$Z(t)V(t) = U(t)S(t), \tag{A.1}$$

where $U(t), S(t), V(t)$ are analytic functions with $U(t), V(t)$ unitary, and $S(t)$ real diagonal.

Proof. The proof is similar to [39, Theorem 1]. [43, pp. 120-122] gives that for some interval around t_0 there exists an analytic eigendecomposition

$$M(t) = \begin{bmatrix} 0 & Z(t) \\ Z(t)^H & 0 \end{bmatrix} = Q(t)\Lambda(t)Q(t)^H, \tag{A.2}$$

with $Q(t)$ unitary and $\Lambda(t)$ real diagonal. The eigenvalues and eigenvectors of $M(t)$ can be paired in the following manner. If $\begin{bmatrix} u(t) & v(t) \end{bmatrix}^H$ is the eigenvector for an eigenvalue $\lambda(t)$, then $\begin{bmatrix} u(t) & -v(t) \end{bmatrix}^H$ is an eigenvector with corresponding eigenvalue $-\lambda(t)$. Thus permuting eq. (A.2) gives

$$Q_1(t) = \frac{1}{\sqrt{2}} \begin{bmatrix} U(t) & U(t) \\ V(t) & -V(t) \end{bmatrix}, \qquad \Lambda_1(t) = \begin{bmatrix} S(t) & 0 \\ 0 & -S(t) \end{bmatrix},$$

such that $M(t) = Q_1(t)\Lambda_1(t)Q_1(t)^H$, where S is real diagonal and U and V are unitary, which implies that $Z(t) = U(t)S(t)V(t)^H$. □

This can not properly be called an analytic SVD, as the singular values are unordered. Still, it is sufficient for the proof of Proposition 5.1.

Data availability

Data will be made available on request.

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